「prop.

Let R be a Commutative Ring with identity.
If every proper ideal in R is prime, then R is field.

Proof. Let a non-zero element $a \in R$ be given.
First, a is not zero divisor, because; suppose that $ax = 0$ for some nonzero $x \in R$.
Then, $(ax) = (0)$ is prime ideal and $ax \in (0)$, but neither a nor x is contained in (0).
(Equivalently, $R/(0) \cong R$ is integral domain, since (0) is prime).

Now, if $(a) = R$, then $ax = 1$ for some x, so a is unit.
If $(a) \neq R$, then $(a^2) \subseteq (a)$, so $(a^2) \neq R$. And, $a^2 \in (a^2)$, so $a \in (a^2)$,
Since a^2 is also prime. Now, $a = a^2 x$ for some $x \in R$, so $1 = ax$. $\boxtimes$